HC-05 Data Sheet

Bluetooth to Serial Port Module

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1、Overview

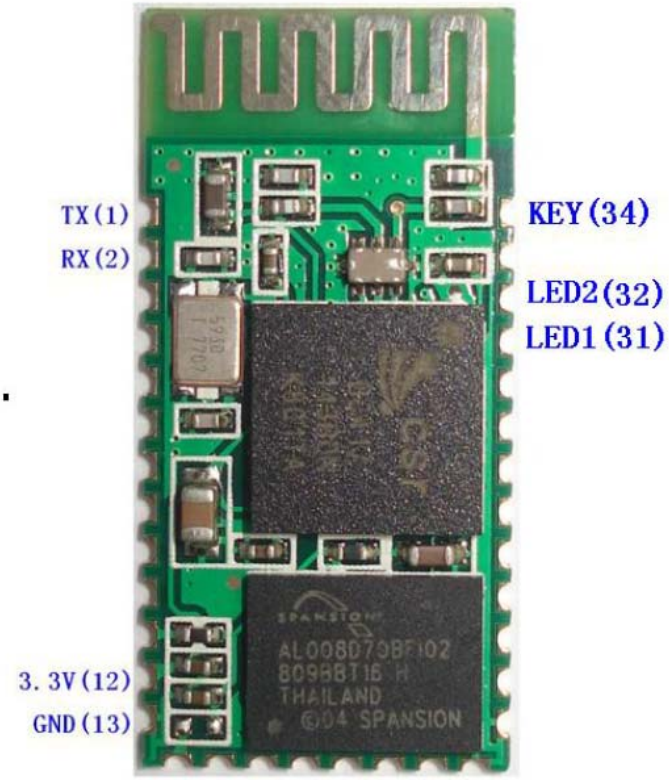
HC-05 module is an easy to use Bluetooth SPP (Serial Port Protocol) module, designed for transparent wireless serial connection setup.

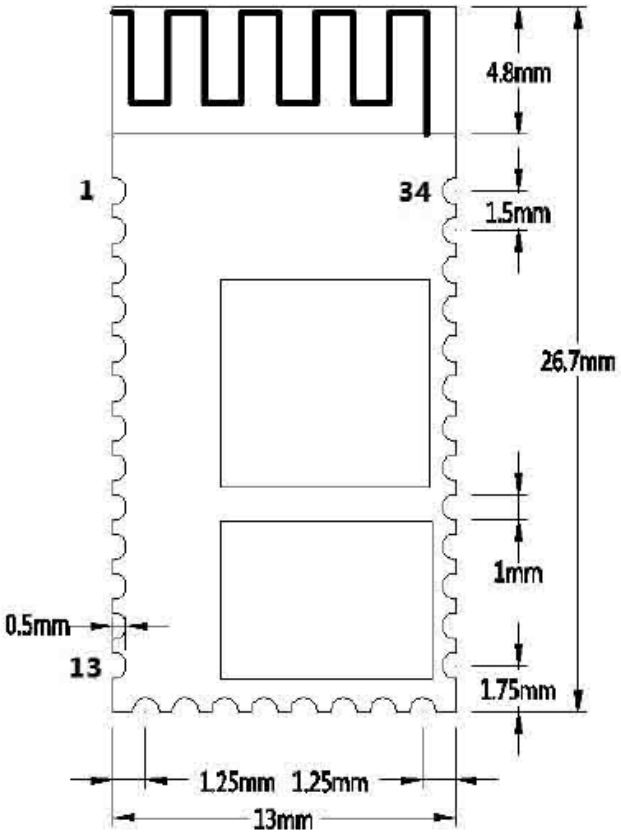
Serial port Bluetooth module is fully qualified Bluetooth V2.0+EDR (Enhanced Data Rate) 3Mbps Modulation with complete 2.4GHz radio transceiver and baseband. It uses CSR Bluecore BC417143 chip. It has the footprint as small as 12.7mmx27mm. Hope it will simplify your overall design/development cycle.

2、Feature

- Sensitivity (Bit error rate) can reach -80dBm,The change range of output's power: -4 - +6dBm.
- Has an EDR module; and the change range of modulation depth: 2Mbps - 3Mbps.
- Has a build-in 2.4GHz antenna; user needn't test antenna.
- Has the external 8Mbit FLASH
- Can work at the low voltage (3.1V~4.2V). The current in pairing is in the range of 30～40mA.
- PIO control can be switched.
- This module can be used in the SMD.
- It's made through RoHS process.
- The board PIN is half hole size.
- Has a 2.4GHz digital wireless transceiver.
- Bases at CSR BC04 Bluetooth technology.
- Has the function of adaptive frequency hopping.
- Small (27mm×13mm×2mm)
- Peripherals circuit is simple.
- It's at the Bluetooth class 2 power level.
- Storage temperature range: -40 ℃ - 85℃， work temperature range: -25 ℃ - +75℃
- Any wave inter Interference: 2.4MHz， the power of emitting: 3 dBm.
- Bit error rate: 0. Only the signal decays at the transmission link, bit error may be produced. For example, when RS232 or TTL is being processed, some signals may decay.

3、Product's picture


TX (1)
RX (2)
KEY (34)
LED2 (32)
LED1 (31)
3. 3V (12)
GND (13)
THAILAND
©04 SPANSION

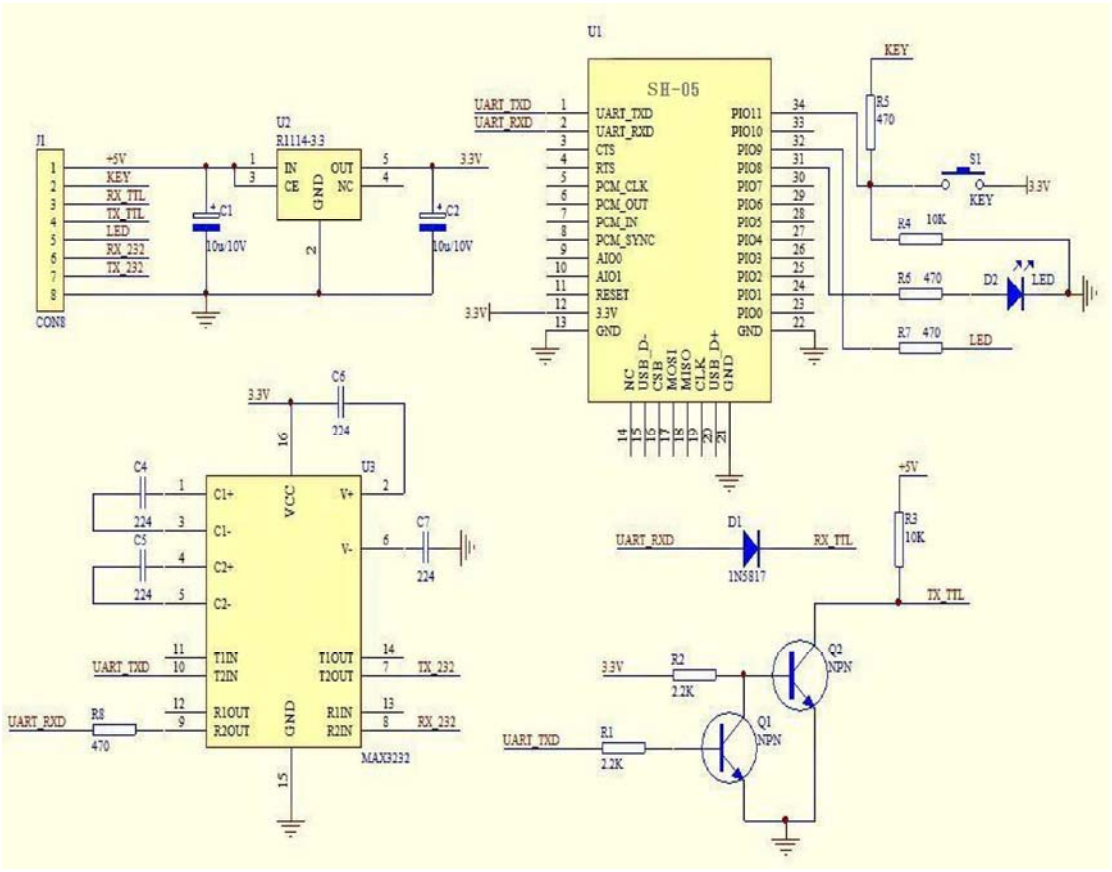

4.8mm
1
34
1.5mm
26.7mm
1mm
0.5mm
13
1.75mm
1.25mm 1.25mm
13mm



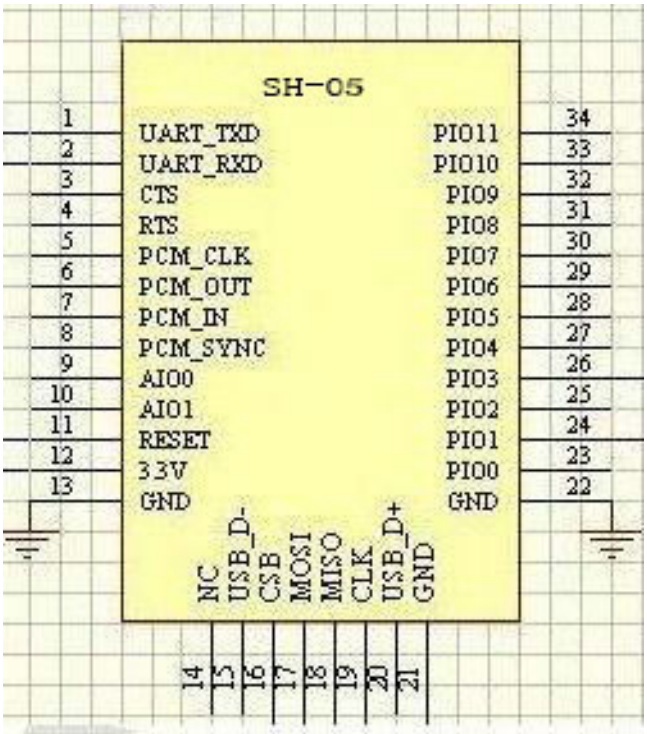
4、Application fields

- Bluetooth Car Handsfree Device
- Bluetooth GPS
- Bluetooth PCMCIA , USB Dongle
- Bluetooth Data Transfer
- Bluetooth Arduino module

5、Block diagram



6、 PINs description



PIN Name	PIN	Pad type	Description	Note
UART_TX	1	CMOS output, Tri-stable with weak internal pull-up	UART Data output	
UART_RX	2	CMOS input with weak internal pull-down	UART Data input	
UART_CTS	3	CMOS input with weak internal pull-down	UART clear to send, active low	
UART_RTS	4	CMOS output, tri-stable with weak internal pull-up	UART r qu st to send, active low	
PCM_CLK	5	Bi-Directional		
PCM_OUT	6	CMOS output		
PCM_IN	7	CMOS Input		
PCM_SYNC	8	Bi-Directional		
AIO0	9	Bi-Directional		
AIO1	10	Bi-Directional		
RESETB	11	CMOS Input with RESETB 11 weak intemal pull-down		

VCC	12	3.3V		
GND	13	VSS	Ground pot	
1V8	14	VDD	Integrated 1.8V (+) supply with On-chip linear regulator output within 1.7-1.9V	
USB_-	15	Bi-Directional		
SPI_CSB	16	CMOS input with weak internal pull-up	Chip select for serial peripheral interface, active low	
SPI_MOSI	17	CMOS input with weak internal pull-down	Serial peripheral interface data input	
SPI_MISO	18	CMOS input with weak internal pull-down	Serial peripheral interface data Output	
SPI_CLK	19	CMOS input with weak internal	Serial peripheral interface clock	
USB_+	20	Bi-Directional		
GND	21	VSS	Ground pot	
GND	22	VSS	Ground pot	
PIO0	23	Bi-Directional RX EN	Programmable input/output line, control output for LNA(if fitted)	

PIO1	24	Bi-Directional TX EN	Programmable input/output line, control output for PA(if fitted)	
PIO2	25	Bi-Directional	Programmable input/output line	
PIO3	26	Bi-Directional	Programmable input/output line	
PIO4	27	Bi-Directional	Programmable input/output line	
PIO5	28	Bi-Directional	Programmable input/output line	
PIO6	29	Bi-Directional	Programmable input/output line	
PIO7	30	Bi-Directional	Programmable input/output line	
PIO8	31	Bi-Directional	Programmable input/output line	LED
PIO9	32	Bi-Directional	Programmable input/output line	LED
PIO10	33	Bi-Directional	Programmable input/output line	
PIO11	34	Bi-Directional	Programmable input/output line	KEY

7、AT Command

More information about command set is provided at hc-05-at-command.pdf.

HC-03/05 Embedded Bluetooth Serial Communication Module

AT command set

Last revised: April, 2011

HC-05 embedded Bluetooth serial communication module (can be short for module) has two work modes: order-response work mode and automatic connection work mode. And there are three work roles (Master, Slave and Loopback) at the automatic connection work mode. When the module is at the automatic connection work mode, it will follow the default way set lastly to transmit the data automatically. When the module is at the order-response work mode, user can send the AT command to the module to set the control parameters and sent control order. The work mode of module can be switched by controlling the module PIN (PIO11) input level.

Serial module PINs:

1. PIO8 connects with LED. When the module is power on, LED will flicker. And the flicker style will indicate which work mode is in using since different mode has different flicker time interval.
2. PIO9 connects with LED. It indicates whether the connection is built or not. When the Bluetooth serial is paired, the LED will be turned on. It means the connection is built successfully.
3. PIO11 is the work mode switch. When this PIN port is input high level, the work mode will become order-response work mode. While this PIN port is input low level or suspended in air, the work mode will become automatic connection work mode.
4. The module can be reset if it is re-powered since there is a reset circuit at the module.

==========Notification ==========

1. How to get to the AT mode.

Way 1:

Step 1: Input low level to PIN34. Step 2: Supply power to the module. Step 3: Input high level to the PIN34. Then the module will enter to AT mode. The baud rate is as

same as the communication time, such as 9600 etc.

Way 2: Step 1: Connect PIN34 to the power supply PIN. Step 2: Supply power to module (the PIN34 is also supplied with high level since the PIN34 is connected with power supply PIN). Then the module will enter to AT module. But at this time, the baud rate is 38400. In this way, user should change the baud rate at the AT mode, if they forget the communication baud rate.

How to get to the communication mode: Step 1: Input low level to PIN34. Step 2: Supply power to the module. Then the module will enter to communication mode. It can be used for pairing.

2. How to set this module be the master role.

Step 1: Input high level to PIO11.

Step 2: Supply power to the module. And the module will enter to the order-response work mode.

Step 3: Set the parameters of the super terminal or the other serial tools (baud rate: 38400, data bit:8, stop bit:1, no parity bit, no Flow Control)

Step 4: Sent the characters “AT+ROLE=1\r\n” through serial, then receive the characters “OK\r\n”. Here, “\r\n” is the CRLF.

Step 5: Input low level to PIO, and supply power to the module again. Then this module will become master role and search the other module (slave role) automatically to build the connection.

3. Notes.

(1) HC-03 and HC-05’s command should end up with “\r\n”. It means when you finish programming, you should add terminator (“ENTER” or “0x0d 0x0a”) to the program. It’s different from HC-04 and HC-06 (They don’t need terminator).

(2) The most common commands for HC-03 and HC-05 are: AT+ROLE (set master –slave), AT+CMODE(set address pairing) , AT+PSWD (set password).

If you want the master module has the function of remembering slave module, the most

simply way is: First, set AT+CMODE=1. Make the master module pair with the slave module. Second, set AT+CMODE=0. Then the master module just can make pair with that specified slave module.

(3) When PIN34 keeps high level, all commands can be used. Otherwise, only some of them can be used.

==========

Detailed description of Command

(AT command is case- sensitive, should end up with terminator (“enter” or “\r\n”).)

1. Test

Command	Response	Parameter
AT	OK	None

2. Reset

Command	Response	Parameter
AT+RESET	OK	None

3. Get the soft version

Command	Response	Parameter
AT+VERSION?	+VERSION: <Param> OK	Param: Version number

Example :

at+version?\r\n

+VERSION:2.0-20100601

OK

4. Restore default status

Command	Response	Parameter
AT+ORGL	OK	None

The parameter of default status:

①. Device type: 0

②. Inquire code: 0x009e8b33

③. Module work mode: Slave Mode

④. Connection mode: Connect to the Bluetooth device specified

⑤. Serial parameter: Baud rate: 38400 bits/s; Stop bit: 1 bit; Parity bit: None.

⑥. Passkey: “1234”

⑦. Device name: “H-C-2010-06-01”

……..

5. Get module Bluetooth address

Command	Response	Parameter
AT+ADDR?	+ADDR: <Param> OK	Param: Bluetooth address

Bluetooth address will show as this way: NAP: UAP: LAP(Hexadecimal)

Example:

Module Bluetooth address: 12: 34: 56: ab: cd: ef

at+addr?\r\n

+ADDR:1234:56:abcdef

OK

6. Set/ inquire device’s name

Command	Response	Parameter
AT+NAME=<Param>	OK	Param: Bluetooth device name Default: “HC-05”
AT+NAME?	1. +NAME:<Param> OK----success 2. FAIL----failure	

Example:

AT+NAME=HC-05\r\n ---set the module device name: “HC-05”

OK

AT+NAME= “HC-05”\r\n ---set the module device name: “HC-05”

OK

at+name=Beijin\r\n ---set the module device name: “Beijin”

OK

at+name= “Beijin”\r\n ---set module device name : “Beijin”

OK

at+name?\r\n

+NAME: Bei jin

OK

7. Get the remote Bluetooth device’s name

Command	Response	Parameter
AT+RNAME?<Param1>	1. +NAME:<Param2> OK----success 2. FAIL----failure	Param1: Remote Bluetooth device address Param2: Remote Bluetooth device address

Bluetooth address will show as this way: NAP:UAP:LAP (Hexadecimal)

Example:

Bluetooth device address: 00:02:72: od: 22 : 24; device name: Bluetooth

at+rname? 0002,72,od2224\r\n

+RNAME:Bluetooth

OK

8. Set/ inquire module role

Command	Response	Parameter
AT+ROLE=<Param>	OK	Param: 0---- Slave role 1---- Master role 2---- Slave-Loop role Default: 0
AT+ ROLE?	+ ROLE:<Param> OK	

Role introduction:

Slave (slave role)----Passive connection;

Slave-Loop----Passive connection, receive the remote Bluetooth master device data and send it back to the master device;

Master (master role)----Inquire the near SPP Bluetooth slave device, build connection with it positively, and build up the transparent data transmission between master and slave device.

9. Set/inquire device type

Command	Response	Parameter
AT+CLASS=<Param>	OK	Param: device type Bluetooth device type is a 32-bit parameter indicates the device type and what type can be supported. Default: 0 More information is provided at the appendix 1(device type introduction).
AT+ CLASS?	1. + CLASS:<Param> OK----success 2. FAIL----failure	

For inquiring the custom Bluetooth device from around Bluetooth devices quickly and effectively, user can set the module to be non-standard Bluetooth device type, such as 0x1f1f (Hexadecimal).

10. Set/ inquire-Inquire access code

Command	Response	Parameter
AT+IAC=<Param>	1. OK----success 2. FAIL----failure	Param: Inquire access code Default: 9e8b33 The more information is provided at the appendix 2(Inquire access code introduction).
AT+ IAC?	+IAC: <Param> OK	

Access code is set to be GIAC type (General Inquire Access Code:0x9e8b33), and used for seeking (or being sought by) all the Bluetooth devices around.

For inquiring (or being inquiring by) the custom Bluetooth device from around Bluetooth devices quickly and effectively, user can set the inquire access code to be the other type number (not GIAC nor LIAC), such as 9e8b3f.

Example:

AT+IAC=9e8b3f\r\n

OK

AT+IAC?\r\n

+IAC: 9e8b3f

OK

11. Set/ inquire - Inquire access mode

Command	Response	Parameter
AT+INQM=<Param>, <Param2>,<Param3>	1. OK----success 2. FAIL----failure	Param: Inquire access mode 0----inquiry_mode_standard 1----inquiry_mode_rssi Param2: the maximum of Bluetooth devices response Param3:The maximum of limited inquiring time The range of limited time: 1~48 (Corresponding time:1.28s~61.44s) Default: 1, 1, 48
AT+ INQM?	+INQM:<Param>,<Param2>,<Param3> OK	

Example:

AT+INQM=1,9,48\r\n ----Set Inquire access mode: 1) has RSSI signal intensity indicator, 2) stop inquiring once more than 9 devices response, 3) limited time is 48*l. 28=61.44s.

OK

AT+INQM\r\n

+INQM:1, 9, 48

OK

12. Set/Inquire- passkey

Command	Response	Parameter
AT+PSWD=<Param>	OK	Param: passkey Default: “1234”
AT+ PSWD?	+ PSWD : <Param> OK	

13.Set/ Inquire- serial parameter

Command	Response	Parameter
AT+UART=<Param>,<Param2>,<Param3>	OK	Param1: baud rate(bits/s) The value (Decimal) should be one of the following: 4800 9600 19200 38400 57600 115200 23400 460800 921600 1382400 Param2:stop bit: 0----1 bit 1----2 bits Param3: parity bit
AT+ UART?	+ UART=<Param>,<Param2>,<Param3> OK	

		0----None 1----Odd parity 2----Even parity Default: 9600, 0, 0

Example:

Set baud rate to be 115200, stop bit to be 2 bits, parity bit to be even parity.

AT+UART=115200,1,2,\r\n

OK

AT+UART?

+UART:115200,1,2

OK

14. Set/ Inquire - connection mode

Command	Response	Parameter
AT+CMODE=<Param>	OK	Param: 0----connect the module to the specified Bluetooth address. (Bluetooth address can be specified by the binding command) 1----connect the module to any address (The specifying address has no effect for this mode.) 2----Slave-Loop Default connection mode: 0
AT+ CMODE?	+ CMODE:<Param> OK	

15. Set/Inquire - bind Bluetooth address

Bluetooth address will show as this way: NAP: UAP:LAP(Hexadecimal)

Command	Response	Parameter
AT+BIND=<Param>	OK	Param----Bluetooth address needed to be bind Default address: 00:00:00:00:00:00
AT+ BIND?	+ BIND:<Param> OK	

Bluetooth address will show as this way: NAP:UAP:LAP(Hexadecimal)

This command is effective only when the module wants to connect to the specified Bluetooth address.

Example:

The module is at connection mode which connects to specified Bluetooth address, and the specified address is 12:34:56:ab:cd:ef.

Command and the response show as follow:

AT+BIND=1234, 56, abcdef\r\n

OK

AT+BIND?\r\n

+BIND:1234:56:abcdef

OK

16. Set/Inquire - drive indication of LED and connection status

Command	Response	Parameter
AT+POLAR=<Param1>, <Param1>	OK	Param1:The value is 0----PI08 outputs low level and turn on LED 1----PI08 outputs high level and turn on LED Param2:The value is 0----PI09 output low level, indicate successful connection 1----PI09 output high level, and
AT+ BIND?	+ POLAR=<Param1>, <Param1> OK	

		indicate successful connection Default: 1, 1

HC-05 Bluetooth module definition: The output of PI08 drives indication of LED work mode; the output of PI09 indicates the connection status.

Example:

PI08 outputs low level and turn on LED, PI09 outputs high level and indicates successful connection.

Command and response show as follow:

AT+POLAR=0, 1\r\n

OK

AT+POLAR?\r\n

+POLAR=0, 1

OK

17. Set PIO single port output

Command	Response	Parameter
AT+PIO=<Param1>,<Param2>	OK	Param1: PIO port number(Decimal) Param2: PIO port status 0----low level 1----high level

HC-05 Bluetooth module provides the user with the ports (PI00~PI07 and PI010) which can extern another input and output ports.

Example:

1. PI010 port outputs high level

AT+PI0=10, 1\r\n

OK

2. PI010 port outpust low level

AT+PI0=10, 0\r\n

OK

18. Set PIO multiple port output

Command	Response	Parameter
AT+MPIO=<Param>	OK	Param: Mask combination of PIO ports number (Decimal)

HC-05 Bluetooth module provides the ports (PI00~PI07 and PI010) which can extern another input and output ports to the user.

(1) Mask of PIO port number = (1<<port number)

(2) Mask combination of PIO ports number= (PIO port number mask 1|PIO port number mask 2|……)

Example :

PI02 port number mask=(1<<2) =0x004

PI010 port number mask =(1<<10)=0x400

Mask combination of PI02 and PI010 port number=(0x004|0x400)=0x404

Example:

1. PI010 and PI02 ports output high level

AT+MPI0=404\r\n

OK

2. PI04 port output high level

AT+PI0=004\r\n

OK

3. PI010 port output high level

AT+PI0=400\r\n

OK

4. All ports output low level

AT+MPI0=0\r\n

OK

19. Inquire PIO port input

Command	Response	Parameter
AT+MPIO?	+MPIO: <Param> OK	Param----PIO port value (16bits) Param[0]=PI00 Param[1]=PI01 Param[2]=PI02 …… Param[10]=PI010 Param[11]=PI011

HC-05 Bluetooth module provides the user with the ports (PI00~PI07 and PI010) which can extern another input and output ports.

20. Set/ Inquire page scan and inquire scan parameter

Command	Response	Parameter
AT+IPSCAN=<Param1>, <Param2>, <Param3>,<Param4>AT+IPSCAN?	OK +IPSCAN: <Param1>,<Param2>, <Param3>,<Param4> OK	Param1:time interval of inquiring Param2: duration in inquiring Param3: time interval of paging Param4: duration in paging The above parameters are decimal. Default:1024,512,1024,512

Example:

at+ipscan=1234,500,1200,250\r\n

OK

at+ipscan?

+IPSCAN:1234,500,1200,250

21. Set/ Inquire—SHIFF energy parameter

Command	Response	Parameter
AT+SNIFF=<Param1>,<Param2>,<Param3>,<Param4>	OK	Param1: maximum time Param2: minimum time Param3: test time Param4: limited time The above parameters are decimal. Default : 0,0,0,0
AT+IPSCAN?	+SNIFF:<Param1>,<Param2>,<Param3>,<Param4>	

22. Set/ Inquire safe and encryption mode

Command	Response	Parameter
AT+SENM=<Param>,<Param2>,	1. OK----success 2. FAIL----failure	Param: the value of safe mode: 0----sec_mode0+off 1----sec_mode1+non_secure 2----sec_mode2_service 3----sec_mode3_link 4----sec_mode_unknown Param2: the value of encryption mode: 0----hci_enc_mode_off 1----hci_enc_mode_pt_to_pt 2----hci_enc_mode_pt_to_pt_and_bcast Default: 0,0
AT+ SENM?	+SENM:<Param>,<Param2>, OK	

23. Delete authenticated device in the Bluetooth pair list

Command	Response	Parameter
AT+PMSAD=<Param>	OK	Param: Bluetooth device address

Example:

Delete the device (address: 12:34:56:ab:cd:ef) in the blue pair list

at+rmsad=1234,56,abcdef\r\n

OK ---- successful deletion

Or

at+rmsad=1234,56,abcdef\r\n

FAIL ----There is no the Bluetooth device whose address is 12:34:56:ab:cd:ef in the pair list.

24. Delete all authenticated devices in the pair list

Command	Response	Parameter
AT+RMAAD	OK	None

Example:

Move all devices away from the pair list.

at+rmaad\r\n

OK

25. Seek the authenticated device in the Bluetooth pair list

Command	Response	Parameter
AT+FSAD=<Param>	1. OK----success 2. FAIL----failure	Param: Bluetooth device address

Example:

Seek the authenticated device (address: 12:34:56:ab:cd:ef) in the pair list

at+fsad=1234,56,abcdef\r\n

OK ----the Bluetooth device whose address is 12:34:56:ab:cd:ef is found.

at+fsad=1234,56,abcde0\r\n

FAIL ----There is no the Bluetooth device whose address is 12:34:56:ab:cd:e0 in the pair list.

26. Get the authenticated device count from the pair list

Command	Response	Parameter

AT+ADCN?	+ADCN:<Param> OK	Param: Authenticated Device Count

Example:

at+adcn?

+ADCN:0 ----There is no authenticated device in the pair list.

OK

27. Get the Bluetooth address of Most Recently Used Authenticated Device

Command	Response	Parameter
AT+MRAD?	+ MRAD : <Param> OK	Param: the Bluetooth address of Most Recently Used Authenticated Device

Example:

at+mrad?

+MRAD:0:0:0 ----There is no device that has been used recently.

OK

28. Get the work status of Bluetooth module

Command	Response	Parameter
AT+STATE?	+ STATE: <Param> OK	Param: work status of module Return value： “INITIALIZED” ----initialized status “READY” ---- ready status “PAIRABLE” ----pairable status “PAIRED” ----paired status “INQUIRING” ----inquiring status “CONNECTING”----connecting status “CONNECTED”----connected status “DISCONNECTED”----disconnected

		status “NUKNOW”----unknown status

Example:

at+state?

+STATE:INITIALIZED ----initialized status

OK

29. Initialize the SPP profile lib

Command	Response	Parameter
AT+INIT	1. OK----success 2. FAIL----failure	None

30. Inquire Bluetooth device

Command	Response	Parameter
AT+INQ	+INQ: <Param1>,<Param2>,<Param3>, …… OK	Param1: Bluetooth address Param2: device type Param3: RSSI signal intensity

Example 1:

at+init\r\n ---- Initialize the SPP profile lib(can’t repeat initialization)

OK

at+iac=9e8b33\r\n ----Inquire Bluetooth device has an access code

OK

at+class=0\r\n ----Inquire the Bluetooth device type

at+inqm=1,9,48\r\n ----Inquire mode: 1) has the RSSI signal intensity indication, 2) stop inquiring if more than 9 Bluetooth devices response, 3) limited time in inquiring is 48*1.28=61.44s.

At+inq\r\n ----inquire the Bluetooth device around

+INQ:2:72:D2224,3E0104,FFBC

+INQ:1234:56:0,1F1F,FFC1

+INQ:1234:56:0,1F1F,FFC0

+INQ:1234:56:0,1F1F,FFC1

+INQ:2:72:D2224,3F0104,FFAD

+INQ:1234:56:0,1F1F,FFBE

+INQ:1234:56:0,1F1F,FFC2

+INQ:1234:56:0,1F1F,FFBE

+INQ:2:72:D2224,3F0104,FFBC

OK

Example 2:

at+iac=9e8b33\r\n ----inquire the Bluetooth device has an access code

OK

at+class=1f1f\r\n ----inquire the Bluetooth device whose device type is 0x1f1f

OK

at+inqm=1,9,48\r\n ----inquire mode: 1) has the RSSI signal intensity indication, 2) stop inquiring if more than 9 Bluetooth devices response, 3) limited time in inquiring is 48*1.28=61.44s

At+inq\r\n ----filter and inquire the Bluetooth device around

+INQ:1234:56:0,1F1F,FFC2

+INQ:1234:56:0,1F1F,FFC1

+INQ:1234:56:0,1F1F,FFC1

+INQ:1234:56:0,1F1F,FFC1

+INQ:1234:56:0,1F1F,FFC2

+INQ:1234:56:0,1F1F,FFC1

+INQ:1234:56:0,1F1F,FFC1

+INQ:1234:56:0,1F1F,FFC0

+INQ:1234:56:0,1F1F,FFC2

OK

Example 3:

at+iac=9e8b3f\r\n ---- inquire the Bluetooth device whose access code is 0x9e8b3f

OK

at+class=1f1f\r\n ----inquire the Bluetooth device whose device type is 0x1f1f

OK

at+inqm=1,1,20\r\n ----inquire mode: 1) Has the RSSI signal intensity indication,
2) stop inquiring if more than 1 Bluetooth device response,
3) limited time in inquiring is 20*1.28=25.6s

At+inq\r\n ----filter and inquire the Bluetooth device around

+INQ:1234:56:ABCDEF,1F1F,FFC2

OK

31. Cancel Bluetooth device

Command	Response	Parameter
AT+INQC	OK	None

32. Set pair

Command	Response	Parameter
AT+PAIR=<Param1>,<Param2>	1. OK----success 2. FAIL----failure	Param1: Bluetooth address of remote device Param2: limited time of connection (second)

Example:

Make pair with the remote Bluetooth device(address:12:34:56:ab:cd:ef), the limited time is 20s.

At+pai=1234,56,abcdef,20\r\n

OK

33. Connect device

Command	Response	Parameter
AT+LINK=<Param>	1. OK----success 2. FAIL----failure	Param: Bluetooth address of remote device

Example:

Connect with the remote Bluetooth device (address: 12:34:56:ab:cd:ef)

at+fsad=1234,56,abcdef\r\n ----To check whether the Bluetooth device (address: 12:34:56:ab:cd:ef) is in the pair list or not.

OK

at+link=1234,56,abcdef\r\n ----The Bluetooth device (address: 12:34:56:ab:cd:ef) is in the pair list. The connection can be built directly without inquiring.

OK

34. Disconnection

Command	Response	Parameter
AT+DISC	1.+DISC:SUCCESS----successful Disconnection OK 2.+DISC:LINK_LOSS----lose the connection OK 3.+DISC:NO_SLC----No SLC connection OK 4、+DISC:TIMEOUT----disconnection timeout OK 5、+DISC:ERROR----disconnection error OK	None

35. Enter to energy mode：

Command	Response	Parameter

AT+ENSNIFF=<Param>	OK	Param: Bluetooth address of device

36. Exit energy mode

Command	Response	Parameter
AT+EXSNIFF=<Param>	OK	Param: Bluetooth address of device

Appendix 1：Introduction of AT command error code

The form of error ---- ERROR:(error_code)

error_code(Hexadecimal)	Note
0	AT command error
1	Default result
2	PSKEY write error
3	Too long length of device name (more than 32 bytes).
4	No device name
5	Bluetooth address: NAP is too long.
6	Bluetooth address: UAP is too long.
7	Bluetooth address: LAP is too long.
8	No PIO number's mask
9	No PIO number
A	No Bluetooth devices.
B	Too length of devices
C	No inquire access code
D	Too long length of inquire access code
E	Invalid inquire access code
F	The length of passkey is 0.
10	Too long length of passkey (more than 16 bytes)
11	Invalid module role
12	Invalid baud rate
13	Invalid stop bit
14	Invalid parity bit
15	Authentication device is not at the pair list.
16	SPP lib hasn't been initialized.
17	SPP lib has been repeated initialization.
18	Invalid inquire mode
19	Too long inquire time
1A	No Bluetooth address
1B	Invalid safe mode
1C	Invalid encryption mode

Appendix 2: The introduction of devices

The Class of Device/Service(CoD) is a 32 bits number that of 3 field specifies the service supported by the device. Another field specifies the minor device class, which describes the device type in more detail

The Class of Device /Service (CoD) field has a variable format. The format is indicated using the ’within the CoD .The length of the Format Type field is variable and ends with two bits different from’11’.The version field starts at the least significant bit of the CoD and may extend upwards. In the ’format#1’ of the CoD (format Type field =00), 11 bits are assigned as a bit –mask (multiple bits can be set) each bit corresponding to a high level generic category of service class. Currently 7 categories are defined. These are primarily of a’ public service’ nature. The remaining 11 bits are used for indicating device type category and other device-specific characteristics. Any reserved but otherwise unassigned bits, such as in the Major Service Class field, should be to 0.

Figure 1.2: The Class of Device/Service field (format type). Please note the krder in which the octets are sent on the air and stored in memory. Bit number 0 is sent first on the air .

1. MAJOR SERVICE CLASSES

Bit no Major Service Class

13 Limited Discoverable Mode [Ref #1]

14 (reserved)

15 (reserved)

16 Positioning(Location identification)

17 Networking (LAN, Ad hoc, …)

18 Rendering (Printing ,Speaker,…)

19 Capturing (Scanner, Microphone,…)

20 0bject Transfer (v-Inbox, v-Folder,…)

21 Audio (Speaker, Microphone, Headset service,…)

22 Telephony (Cordless telephony, Modem, Headset service,…)

23 Information (WEB-server, WAP- server,…)

TABLE 1.2:MAJOR SERVICE CLASSES

[Ref #1 As defined in See Generic Access Profile, Bluetooth SIG]

2. MAJOR DEVICE CLASSES

The Major Class segment is the highest level of granularity for defining a Bluetooth Device. The main function of a device is used for determining the major Class grouping. There are 32 different possible major classes. The assignment of this Major Class field is defined in Table1.3.

1 2 1 1 1 0 9 8 Major Device Class

0 0 0 0 0 Miscel laneous [Ref #2]

0 0 0 0 1 Computer (desktop, notebook, PDA, organizers,…)

0 0 0 1 0 Phone (cellular ,cordless ,payphone, modem,…)

0 0 0 1 1 LAN/Network Access point

0 0 1 0 0 Audio/Video (headset, speaker, stereo, video display, vcr …)

0 0 1 0 1 Periphereal (mouse, joystick, keyboards.…)

0 0 1 1 0 Imaging (printing, scanner, camera, display,…)

1 1 1 1 1 Uncategorized, specific device code not specified

X X X X All other values reserved

TABLE 1.3: MAJOE DEVICE CLASSES

[Ref #2:Used where a more specific Major Device Class is not suited (but only as specified as in this document) .Devices that do not have a major class assigned can use the all-1 code until' classified']

3. THE MINOR DEVICE CLASS FIELD

The' Minor Device Class field' (bits 7 to 2 in the CoD), are to be interpreted only in the context of the Major Device Class (but interpreted of the Service Class field). Thus the meaning of the bits may change, depending on the value of the ' Major Device Class field'. When the Minor Device Class field indicates a device class ,then the

primary decvice class should be reported, e. g . a cellular phone that can work as a cordless handset should

4. MINOR DEVICE CLASS FIELD–COMPUTER MAJOR CLASS

Minor Device Class

7 6 5 4 3 2 bit no of CoD

0 0 0 0 0 0 Uncategorized, code for device not assigned

0 0 0 0 0 1 Desktop workstation

0 0 0 0 1 0 Server-class computer

0 0 0 0 1 1 Laptop

0 0 0 1 0 0 Handheld PC/PDA(clam shell)

0 0 0 1 0 1 Palm sized PC/PDA

0 0 0 1 1 0 Wearable computer (Watch sized)

X X X X X X All other values reserved

TABLE 1.4: SUB DEVICE CLASS FIELD FOR THE' COMPUTER 'MAJOR CLASS

5. MINOR DEVICE CLASS FIELD – PHONE MAJOR CLASS

Minor Device Class

7 6 5 4 3 2 bit no of CoD

0 0 0 0 0 0 Uncategorized, code for device not assigned

000001 Cellular

0 0 0 0 1 0 Cordless

0 0 0 0 1 1 Smart phone

0 0 0 1 0 0 Wired modem or voice gateway

0 0 0 1 0 1 Common ISDN Access

0 0 0 1 1 0 Sim Card Reader

X X X X X X All other values reserved

TABLE1.5: SUB DEVICE CLASSES FOR THE'PHONE' MAJOR CLASS

6. MINOR DEVICE CLASS FIELD –LAN/NETWORK ACCESS POINE MAJOR

CLASS

Minor Device Class

7 6 5 bit no of CoD

0 0 0 Fully available

0 0 1 1 – 17% utilized

0 1 0 1 7 - 33% utilized

0 1 1 3 3 – 50% utilized

1 0 0 5 0 – 67% utilized

1 0 1 6 7 – 83% utilized

1 1 0 8 3 – 99% utilized

1 1 1 No service available [REF #3]

XXX All other values reserved

TABLE1.6: THE LAN/NETWORK ACCESS POINE LOAD FACTOR FIELD

[Ref #3:“Device is fully utilized and cannot accept additional connections at this time, please retry later”]

The exact loading formula is not standardized. It is up to each LAN/Network Access Point implementation to determine what internal conditions to report as a utilization of communication requirement is that the box .As a recommendation, a client that locates multiple LAN/Network Access Points should attempt to connect to the one reporting the lowest load.

Minor Device Class

4 3 2 bit no of CoD

0 0 0 Uncategorized (use this value if no other apply)

XXX All other values reserved

TABLE1.7:RESERVED SUB-FIELD FOR THE LAN/NETWORK ACCESS POINE

7. MINOR DEVICE CLASS FIELD – AUDIO/VIDEO MAJOR CLASS

Minor Device Class

7 6 5 4 3 2 bit no of CoD

0 0 0 0 0 0 Uncategorized, code not assigned

0 0 0 0 0 1 Device conforms to the Headset profile

000010 Hands-free

0 0 0 0 1 1 (Reserved)

0 0 0 1 0 0 Microphone

0 0 0 1 0 1 Loudspeaker

0 0 0 1 1 0 Headphones

0 0 0 1 1 1 Portable Audio

0 0 1 0 0 0 Car audio

0 0 1 0 0 1 Set-top box

0 0 1 0 1 0 HiFi Audio Device

001011 VCR

0 0 1 1 0 1 Camcorder

0 0 1 1 1 0 Video Monitor

0 0 1 1 1 1 Video Display and Loudspeaker

0 1 0 0 0 0 Video Conferencing

0 1 0 0 0 1 (Reserved)

0 1 0 0 1 0 Gaming/Toy [Ref #4]

X X X X X X All other values reserved

[Ret #4: Only to be used with a Gaming/Toy device that makes audio/video capabilities available via Bluetooth]

TABLE 1.8: SUB DEVICES FOR THE ’AUDIO/VIOEO’MAJOR CLASS

8. MINOR DEVICE CLASS FIELD – PERIPHERAL MAJOR CLASS

Minor Device Class

7 6 bit no of CoD

0 1 Keyboard

1 0 Pointing device

1 1 Combo keyboard /pointing device

X X X All other values reserved

TABLE1.9: THE PERIPHERAL MAJOR CLASS KEYBOARD/POINTING DEVICE

FIELD

Bits 6 and 7 independently specify mouse, keyboard or combo mouse/keyboard devices. These may be combined with the lower bits in a multifunctional device.

Minor Device Class

5 4 3 2 bit no of CoD

0 0 0 0 Uncategorized device

0 0 0 1 Gamepd

0 0 1 1 Remote control

0 1 0 0 Sensing device

0 1 0 1 Digitizer tablet

X X X X All other values reserved

TABLE1.10: RESERVED SUB-FIELD FOR THE DEVICE TYPE

9. MINOR DEVICE CLASS FIELD – IMAGING MAJOR CLASS

Minor Device Class

7 6 5 4 bit no of CoD

X X X 1 Display

X X 1 X Camera

X 1 X X Scanner

1 X X X Printer

X X X X All other values reserved

TABLE 1.11: THE TMAGING MAJOR CLASS BITS 7 TO 7

Bits 4 to 7 independently specify bi splay, camera, scanner or printer. These may be combined in a multifunctional device.

Minor Device Class

3 2 bit no of CoD

0 0 Uncategorized, default

X X All other values reserved

TABLE 1. 12: THE IMAGING MAJOR CLASS BITS 2 AND 3

Bits 2 and 3 are reserved

Appendix 3: (The Inquiry Access Codes)

The General-and Device-Specific Inquiry Access Codes (DIACs)

The Inquiry Access Code is the first level of filtering when finding Bluetooth devices. The main purpose of defining multiple IACs is to limit the number of responses that are received when scanning devices within range.

0. 0x9E8B33 ---- General/Unlimited Inquiry Access Code(GIAC)
1. 0x9E8B00 ---- Limited Dedicated Inquiry Access Code(LIAC)
2. 0x9E8B01 ～ 0x9E8B32 RESERVED FOR FUTURE USE
3. 0x9E8B34 ～ 0x9E8B3F RESERVED FOR FUTURE USE

The Limited Inquiry Access Code(LIAC)is only intended to be used for limited time periods in scenarios where both sides have been explicitly caused to enter this state, usually by user action. For further explanation of the use of the LIAC, please refer to the Generic Access Profile.

In contrast it is allowed to be continuously scanning for the General Inquiry Access Code (GIAC)and respond whenever inquired.

Dear customer,

Regarding your issue, please refer the following:

Q001	Debugging tools and where to download them?
	Press the button, Go to "AT" mode. Sending "AT" (ending on a new line) will get "OK" through the serial port. If you have problems entering the AT command mode, we recommend downloading D SD TECH Bluetooth Tools to test it. Using this software, you can test in command mode and also set some paramete rs such as baud rate without having to memorize the AT command. You can download the software from ： https://www.deshide.com/News-detail_DSDTechTools.html or Google search for "DSD TEH Bluetooth Tools".

Q002	The logic level of this module is 3.3V (for tx, rx), how do I connect the module if my board is 5v?
	If your board has a logic level of 5V, then a voltage divider must be added to the arduino Tx

Q003	What is the default serial port setting?
	The default baud rate is set to 9600 N 8 1

Q004	How do I connect the module to the MCU or Arduino?
	Bluetooth module tx -> Arduino/MCU rx Bluetooth module rx -> Arduino/MCU tx.

Q005	What should I do if I have been unable to send or receive data after connecting?
	If the serial port does not respond, switch TX and RX.

Q006	What is the correct format of AT commands?
	AT commands should end on a new line.

Q007	What are the characteristics of Bluetooth module responding to AT commands?
	The Bluetooth module only responds to AT commands sent from the serial ports (TX, RX). It does not respond to AT commands sent via Bluetooth.

Q008	How to physically connect the HC-05 adapter to a PC？
	You can use a USB to TTL adapter to make a physical connection. TXD -> RXD RXD -> TXD VCC -> VCC GND -> GND

Q009	How do I put this module in a state where I can connect to it using Windows?
	After powering up the Bluetooth module, open the Windows Bluetooth to connect, and a virtual COM port will appear after connection.

Q010	Why has been the module not responding to any AT commands?
	To enter the AT command mode of HC-05, you need to press the button on the module, you can try to use a USB to TTL adapter to connect directly to the HC-05 module to troubleshoot the Arduino serial port parameter configuration.

Q011	How is this product wired?
	For wiring please refer to. VCC -> 5V GND -> GND TXD -> RXD-0 RXD -> TXD-1

Q012	How do I determine if the Bluetooth module is damaged?
	1. You can try to connect the Bluetooth module directly to the USB to TTL adapter, and the Android phone is connected to the Bluetooth for sending and receiving data test. 2、You can also try to light up the LED on the arudino board first.

Q013	How do I connect the Bluetooth module from a Windows computer?
	HC-05,HC-06 are normally connected to Windows, but does not support ios system, you can directly to the bluetooth module power supply, open the windows bluetooth to connect, after connecting the virtual COM port will appear, you can view in the device.

Q014	How do I put the module in a state where I can connect to it using Windows?
	After powering up the Bluetooth module, open the Bluetooth of Windows to connect, after connecting, the virtual COM port will appear.

Q015	How to configure the baud rate of this module?

	Please refer to the picture for the baud rate setting.

13.Set/ Inquire- serial parameter

Command	Response	Parameter
AT+UART=<Param>,<Param2>,<Param3>	OK	Param1: baud rate(bits/s) The value (Decimal) should be one of the following: 4800 9600 19200 38400 57600 115200 23400 460800 921600 1382400 Param2:stop bit: 0----1 bit 1----2 bits Param3: parity bit
AT+ UART?	+ UART=<Param>,<Param2>,<Param3> OK	
		0----None 1----Odd parity 2----Even parity Default: 9600, 0, 0

Example:

Set baud rate to be 115200, stop bit to be 2 bits, parity bit to be even parity.

AT+UART=115200,1,2,\r\n

OK

AT+UART?

+UART:115200,1,2

OK

14. Set/ Inquire - connection mode

Command	Response	Parameter
AT+CMODE=<Param>	OK	Param: 0----connect the module to the specified Bluetooth address. (Bluetooth address can be specified by the binding command) 1----connect the module to any address (The specifying address has no effect for this mode.) 2----Slave-Loop Default connection mode: 0
AT+ CMODE?	+ CMODE:<Param> OK	

Q016	How to enter AT command mode?
	power you hc-05 module , you only need to press the button on HC-05 ,hc05 is enter the AT command mode. and the baud rate is 9600.

Q017	I followed the instructions to connect to the arduino, but the Windows 11 laptop will not pair with the HC-05. The computer was able to discover the device, but it would ask me for a password and then would not connect. Used the default 1234 password provided. But the connection fails.
	After entering the password, the module light flashes slowly indicating that the bluetooth is successfully connected, but the bluetooth panel of windows 11 will show that it is not connected. At this time there will be two Bluetooth serial ports on the computer, through these two serial ports can realize the Bluetooth serial communication.

Q018	How is the hc-05 module indicated by the led connection indicator?
	In the event that the HC-05 module is powered on but not connected for pairing, the LED indicator will flash rapidly to indicate that the HC-05 module is waiting for a pairing connection. Once the HC-05 module has been successfully paired and connected to a device, either as a master or a slave, the LED indicator's blinking pattern will change to two blinks and a stop, a specific blinking pattern that clearly indicates that the module is currently connected.

If you still have problem, please tell us.

DSD TECH team is always there.

Our email: dsd_tech@outlook.com

website: http://www.deshide.com

Regards

DSD TECH Team

2024/11/29

EGBT-045MS
EGBT-046S

Bluetooth Modules

Wireless UART Cable Replacement

Hardware Manual &
AT Commands Reference Manual Rev. 1r0

EGBT-045MS and EGBT-046S Bluetooth Module are low cost replacements of our now retired EGBC-04 Bluetooth Module. EGBC-04 is an excellent Bluetooth Module, it is fully certified to Bluetooth standards, and is loaded with programmable features users had come to love. There is just one thing that went against it- it is expensive.

It is easy to see why the EGBC-04 cost so much. Firstly, the manufacturer produced these specialty modules in relatively small volume; hence, there is no economy of the scale to speak of. Secondly, certification costs a lot of money; and this cost will have to be added on top of the manufacturing cost. Hence, EGBC-04 ended up costing about 10 times more expensive than its garden variety USB-type Bluetooth dongles cousins.

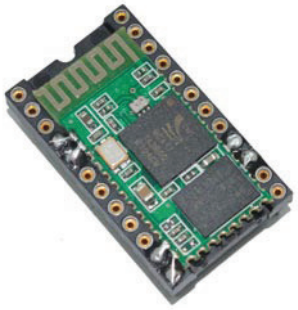
Fortunately, at least one volume manufacturer have came up with an idea of producing a generic Bluetooth module in large quantity, for sale and distribution to developers who now have to put only the firmware functionalities. This resulted in a huge drop in prices of these specialty Bluetooth modules, benefiting us experimenters and hobbyists.

EGBT-045MS and EGBT-046S are generic Bluetooth Modules loaded with SPP firmware for UART wireless cable replacement functions. The EGBT-045MS can be configured by the user to work either as a master or slave Bluetooth device using a set of AT commands.

EGBT-046S, on the other hand, is permanently programmed as Bluetooth slave device. EGBT-046S, because of its simpler function, is a lot easier to use, and of course, costs less than EGBT-045MS. You can use it straight out of the box as a UART wireless cable replacement, without any need to add set-up codes in your microcontroller application firmware.



The new EGBT-04 Bluetooth module comes in two flavors. The EGBT-046S is permanently configured as a slave device. EGBT-045MS, on the other hand, can be configured by the user to work as a master or slave Bluetooth device.



EGBT-04 modules can be soldered directly on a hi-rel type IC socket to make it easier to work with prototyping platforms, such as breadboards and perforated prototyping boards.

Use the cheaper EGBT-046S if your application will connect to a master Bluetooth device, such as PC or laptops. Use the EGBT-045MS if your application must connect to a slave Bluetooth device, such as with EGBT-046S. Note that EGBT-045MS will work as well as a slave Bluetooth device.

COMMON SPECIFICATIONS

Radio Chip: CSR BC417
Memory: External 8Mbit Flash
Output Power: -4 to +6dbm Class 2
Sensitivity: -80dbm Typical
Bit Rate: EDR, up to 3Mbps
Interface: UART
Antenna: Built-in
Dimension: 27W x 13H mm

Voltage: 3.1 to 4.2VDC
Current: 40mA max

COMMON HARDWARE INTERFACING CONSIDERATIONS

The EGBT-04 module will work with supply voltage of 3.1VDC to 4.2VDC. When supplied with 3.3VDC, it will interface directly with the UART port of any microcontroller chip running at 3.3VDC.

When used with 5V microcontrollers, The TXD output logic swing of the EGBT-04 still falls within the valid 5V TTL range, hence, can be connected directly to the UART RXD of the 5V microcontroller host. EGBT RXD and inputs, however, are not 5V tolerant, and can be damaged by 5V level logic going in. Some level translation circuit must be added to protect the inputs.

A simple diode level translator circuit like the ones shown in Figure 3 and 7 will suffice in most applications. A better alternative is with the use of 5V input tolerant tiny logic chips such as 74LVC1G125 – a single buffer chip housed in smd sot23-5 package.

EGBT-046S PIN CONFIGURATION

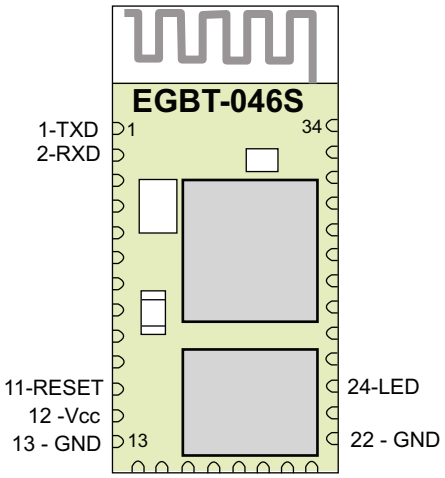


Figure 1. *EGBT-046S Pin Layout*

Table 1. EGBT-046S Pin Description

PIN	ID	DESCRIPTION
1	TXD	UART TXD Output
2	RXD	UART RXD Input
11	RESET	RESET Input
12	Vcc	+3.1 to 4.2VDC Power Input
13	GND	Common Ground
22	GND	Common Ground
24	LED	LED Status Indicator Flashing - Waiting to Connect/Pair Steady ON - Connected/Paired

Note:
All unassigned pins must be left unconnected.

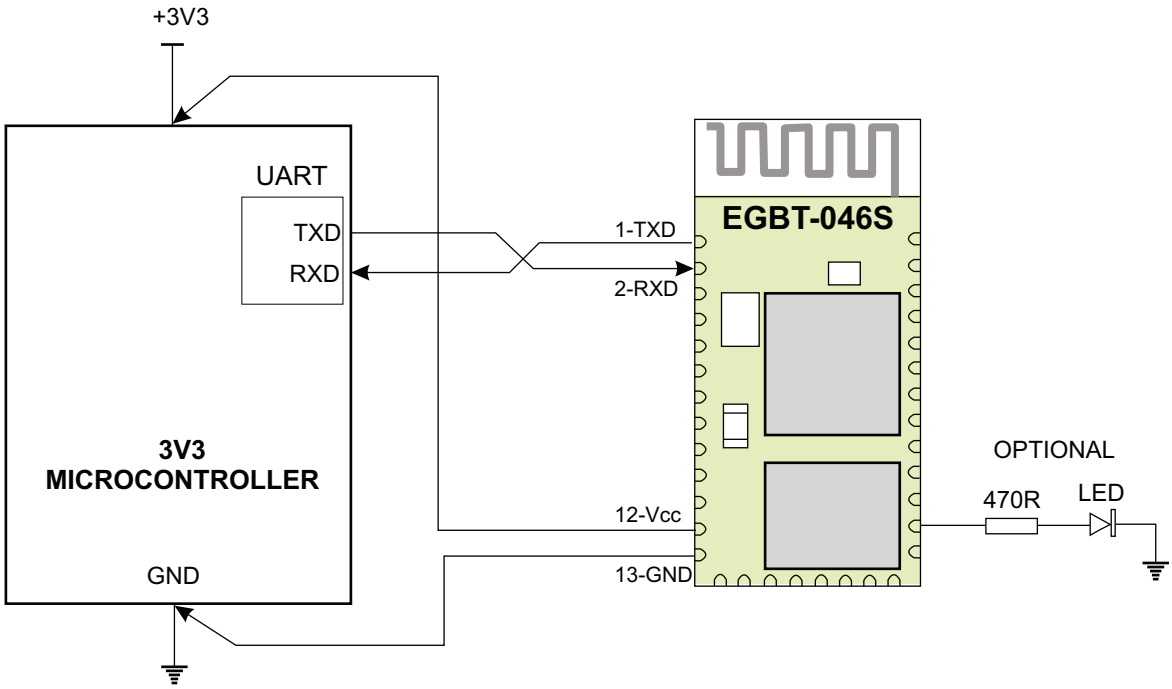


Figure 2. *EGBT-046S wiring example with a 3v3 host microcontroller. The 470R resistor and LED are for status indication, and may be omitted if not needed.*

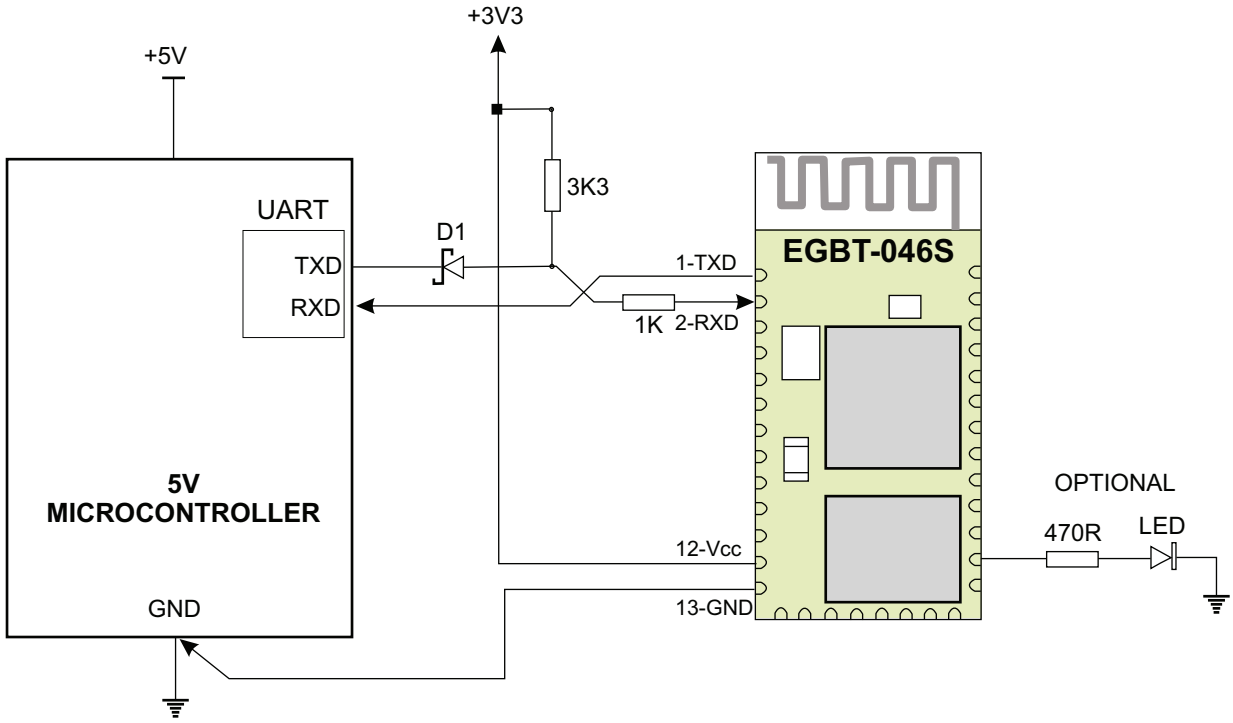


Figure 3. *EGBT-046S RX input is not 5V tolerant. A schottky diode connected as shown will keep 5V voltages out of the Bluetooth module when operated with a 5V host microcontroller.*

PREPARATION FOR USE

The EGBT-046S is permanently configured as a slave Bluetooth device. It works under the following default configuration:

Baud Rate: 9600 bps
Data : 8 bits
Stop Bits: 1 bit
Parity : None
Handshake: None

Passkey: 1234
Device Name: linvor

If the default configuration suits your application, then you can use EGBT-046S immediately. Once it is paired to a master Bluetooth device, its operation becomes transparent to the user. No user code specific to the Bluetooth module is needed at all in the user microcontroller program.

The EGBT-046S automatically sets itself up in Command Mode when it is not remotely connected (paired) to any other Bluetooth device. You can change the Passkey, Device Name, and Baud Rate while the EGBT-046S is in Command Mode by entering a small subset of AT style commands. Any changes made will be retained even after power is removed from the EGBT-046S, hence device configuration setup must not be repeated unless new changes need to be made.

You can do configuration setup using the host controller itself (the microcontroller in your own circuit), or a PC running a terminal software using a serial to TTL (or USB to Serial TTL) converter. See Figure 4 for connection details.

It is important to note that EGBT-046S does not wait for any termination character for each AT command entry. Instead, it acts to whatever character you entered after one second. Hence, if you are not able to complete a command entry within a second, it will be ignored. Because of this behavior, it may be extremely difficult to do manual entry configuration using Windows Hyperterminal software. Terminal software that allows batch sending of multiple characters must be used.

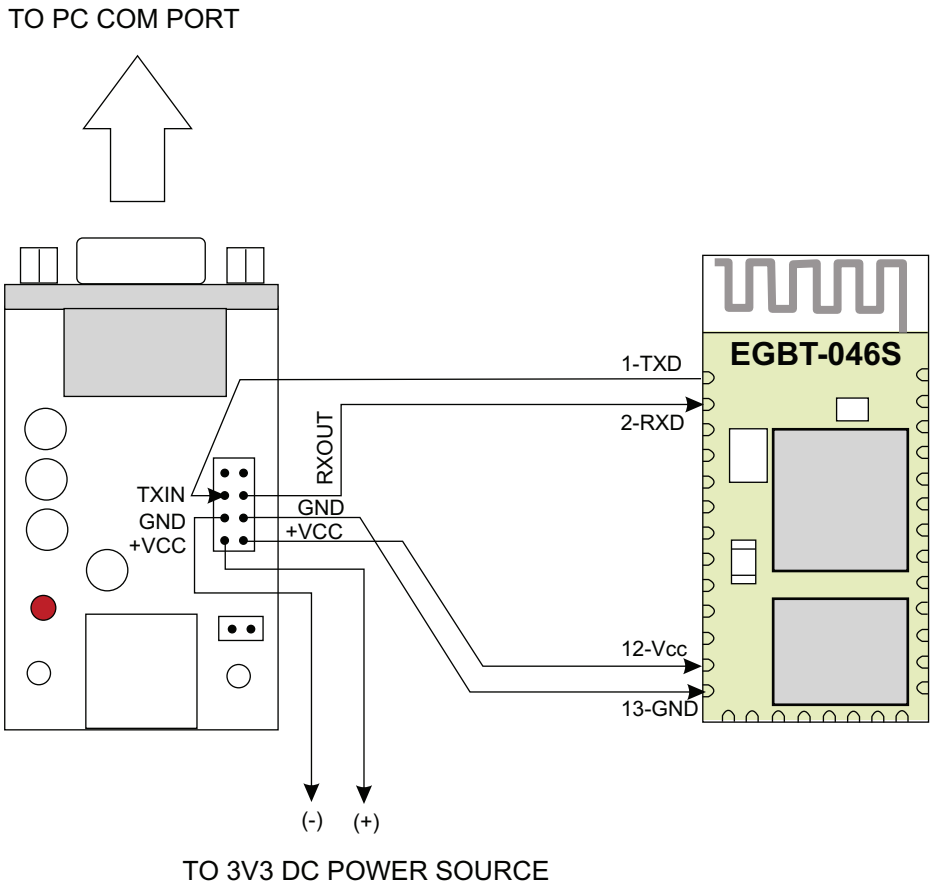


Figure 4. *A PC may be used to configure the EGBT-04. To connect to a PC COM port, a RS-232C to TTL converter is needed. This figure shows a wiring example using e-Gizmo RS-232 to TTL converter kit.*

EGBT-046S AT Command Set

1. TEST

Used to test the UART connection between the host controller and Bluetooth Module.

COMMAND	RESPONSE
AT	OK

2. Change Baud Rate

COMMAND	RESPONSE
AT+BAUD<p>	OK<r>

where:

<p> Paramater
<r> Response, set to nnnn bps

<p>	<r>	Remarks
1	1200	set to 1200bps
2	2400	set to 2400bps
3	4800	set to 4800bps
4	9600	set to 9600bps (Default)
5	19200	set to 19200bps
6	38400	set to 38400bps
7	57600	set to 57600bps
8	115200	set to 115200bps
9	230400	set to 230400bps
A	460800	set to 460800bps
B	921600	set to 921600bps
C	1382400	set to 1382400bps

Caution:
PC standard COM port hardware does not support baud rates in excess of 115200bps. If you are using a PC to configure EGBT-046S and accidentally set baud rate to these values, connection to a PC COM port will no longer be possible. Use of USB to Serial converter cable that can work at higher bauds may be necessary to re-establish a connection. Prolific PL-2303 based USB to Serial converter cables are known to work up to 921600bps.

Example1: Set baud rate to 57600bps

From Host controller:
AT+BAUD7
EGBT-046S Response
OK57600

Example2: Set baud rate to 4800bps

From Host controller:
AT+BAUD3
EGBT-046S Response
OK4800

3. Change Device Name

The EGBT-046S can be assigned a readable name of up to 20 characters in lenght.

COMMAND	RESPONSE
AT+NAME<name>	OK<name>

Example1: Set device name as EGBT-04

From Host controller:
AT+NAMEEGBT-04
EGBT-046S Response
OKEGBT-04

4. Change PASSKEY(PIN code)

Passkey (PIN Code) is a 4-digit code shared with a master Bluetooth Device (e.g. PC) to prevent unauthorized pairing.

COMMAND	RESPONSE
AT+PIN<nnnn>	OK<nnnn>

Where:
<nnnn> 4-digit passkey

Example1: Set PASSKEY to 5995

From Host controller:
AT+PIN5995
EGBT-046S Response
OK5995

EGBT-045MS PIN CONFIGURATION

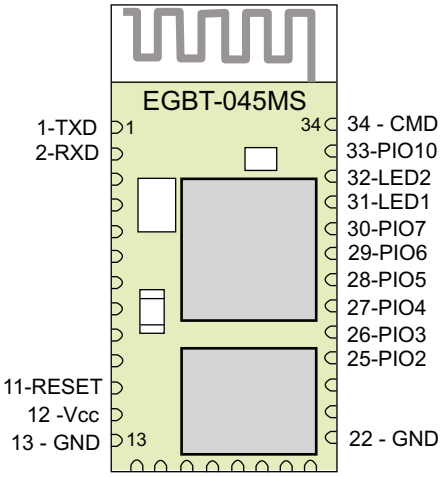


Figure 5. *EGBT-045MS pin layout*

Table 1. EGBT-045MS Pin Description

PIN	ID	DESCRIPTION
1	TXD	UART TXD Output
2	RXD	UART RXD Input
11	RESET	RESET Input
12	Vcc	+3.1 to 4.2VDC Power Input
13	GND	Common Ground
22	GND	Common Ground
25	PIO2	User programmable I/O
26	PIO3	User programmable I/O
27	PIO4	User programmable I/O
28	PIO5	User programmable I/O
29	PIO6	User programmable I/O
30	PIO7	User programmable I/O
33	PIO10	User programmable I/O
31	LED1	LED Status Indicator
32	LED2	LED Status Indicator
34	CMD	Command Mode

Note:
All unassigned pins must be left unconnected.

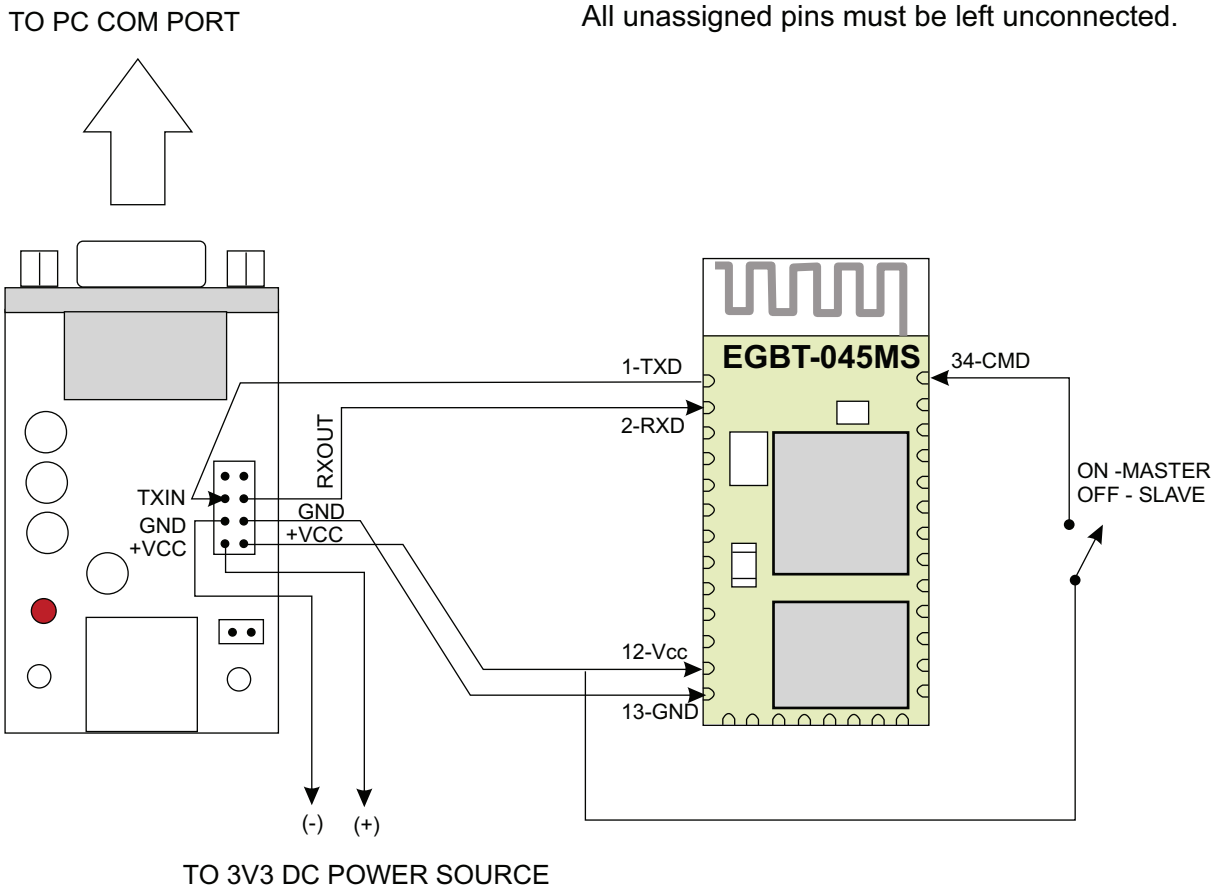


Figure 6. *Connecting the EGBT-045MS to a PC for test and configuration. To connect to a PC COM port, a RS-232C to TTL converter is needed. This figure shows a wiring example using e-Gizmo RS-232 to TTL converter kit.*

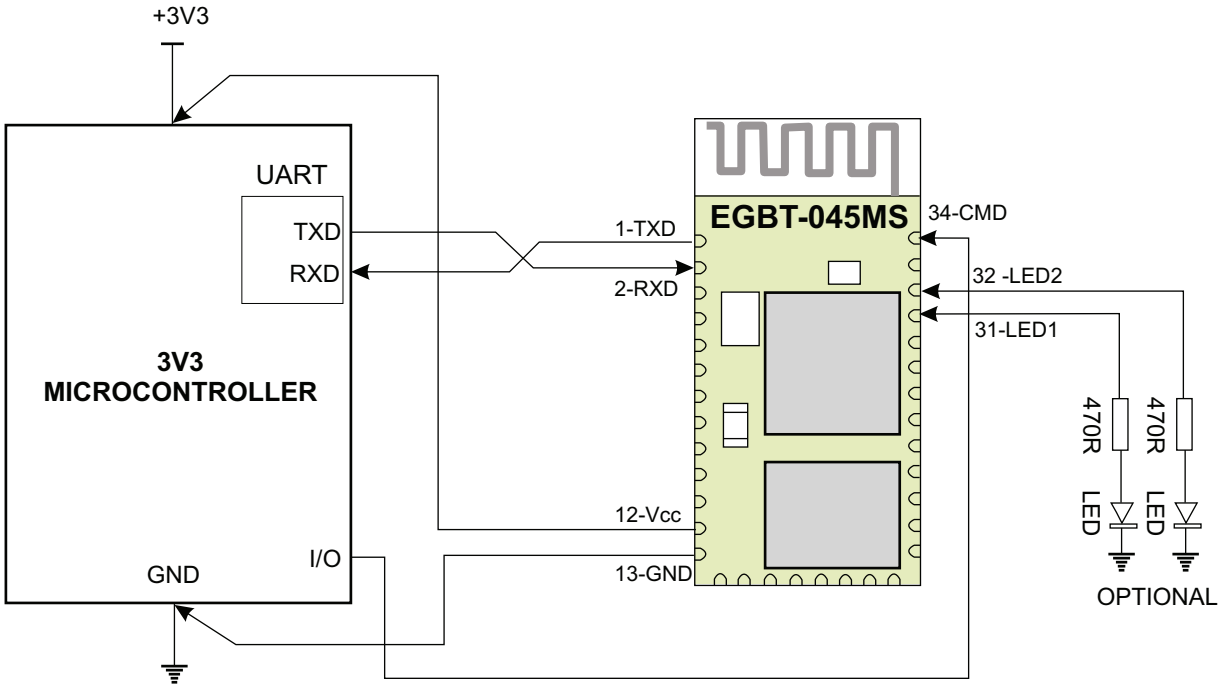


Figure 7. *EGBT-045MS wiring example with a 3v3 host microcontroller. The 470R resistors and LEDs are for status indication, and may be omitted if not needed. The microcontroller can switch the EGBT-045MS between command mode and data mode at any time by setting the CMD pin accordingly.*

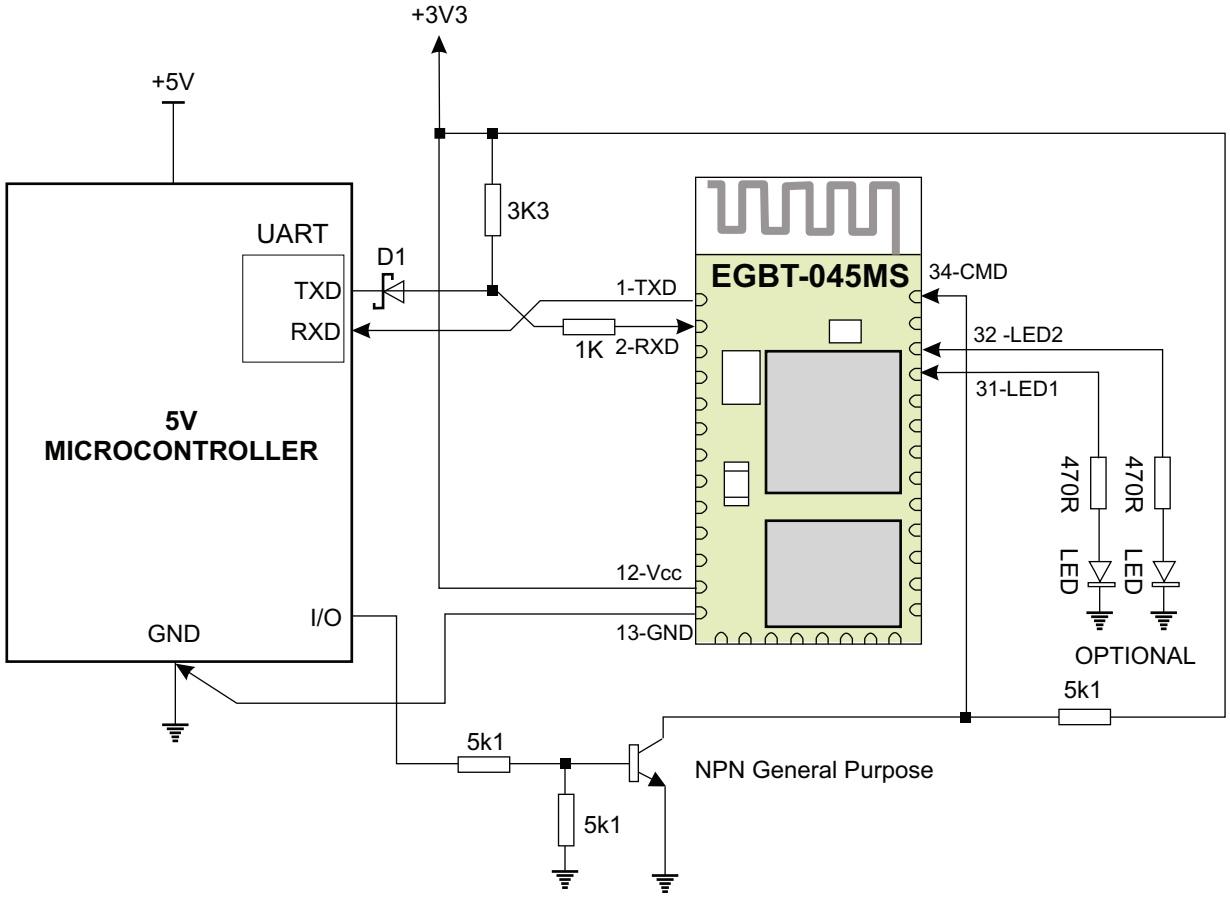


Figure 8. *EGBT-045MS RX input is not 5V tolerant. A schottky diode connected as shown will keep 5V voltages out of the Bluetooth module when operated with a 5V host microcontroller. The transistor circuit takes on a similar function for the CMD pin.*

PREPARATION FOR USE

The EGBT-045MS is pre-configured as a slave Bluetooth device. As shipped, it works under the following data mode default configuration:

Baud Rate: 38400/9600 bps – see “Entering Command Mode” section for details
Data : 8 bits
Stop Bits: 1 bit
Parity : None
Handshake: None

Passkey: 1234
Device Name: HC-05

If the default configuration suits your application, then you can use EGBT-045MS immediately. Once it is paired to a master Bluetooth device, its operation becomes transparent to the user. No user code specific to the Bluetooth module is needed at all in the user microcontroller program.

The EGBT-045MS works in Data Mode when the CMD pin (pin 34) is pulled to logic 0 level or is left unconnected. Transparent UART data transfer with a connected remote device occurs only while in Data Mode. The EGBT-045MS switches to Command Mode if CMD pin is set to logic HIGH. Command mode must be invoked to setup and configure EGBT-045MS. Pairing and connection to a remote Bluetooth slave device can be initiated only while the EGBT-045MS is in command mode (and configured in role as a Master) by entering a string of AT style commands. Any changes made to system parameters (e.g. password, baud rate, etc) will be retained even after power is removed, hence device configuration setup must not be repeated unless new changes need to be made.

Entering Command Mode

EGBT-045MS is put in Command Mode by setting the CMD pin to logic High. There is one important point the user should be aware of when setting the device in Command Mode – the baud rate may assume a different value depending on the instance the CMD pin is switched to high:

• If EGBT-045MS is powered ON with CMD pin to +Vcc, the UART is set to 38400bps, 8 data bits, 1 stop bit, no parity, no handshake.

• If the EGBT-045MS is powered ON with the CMD pin at logic low (or open circuit), and then pulled High a moment later, the UART parameters assumes a set of values that was previously fixed using the AT+UART command, with 9600bps, 8 data bits, 1 stop bit, no parity, and no handshake as default parameter values.

You can do configuration setup using the host controller itself (the microcontroller in your own circuit), or a PC running a terminal software using a serial to TTL (or USB to Serial TTL) converter. See Figure 6 for connection details.

EGBT-045MS AT COMMAND REFERENCE

Symbols and Nomenclatures.

<xxxx> Descriptive ID of parameters that must be entered by the user or reported back by the EGBT-045MS.

Example:
AT+INQM=<inq1>,<inq2>,<inq3>
May in actual use appear as
AT+INQM=0,1,15
When <inq1> =0, <inq2>=1, and <inq3>=15

↵ = Carriage Return followed by Line Feed character<CR><LF>

Example:

OK↵ = OK<CR><LF>

Important note: All commands must be terminated by <CR><LF>. If the host controller send a <CR> only, EGBT-045MS will repeatedly send a respond that will stop only when <LF> is issued by the host controller

AT COMMAND LISTING

	COMMAND	FUNCTION
1	AT	Test UART Connection
2	AT+RESET	Reset Device
3	AT+VERSION	Querry firmware version
4	AT+ORGL	Restore settings to Factory Defaults
5	AT+ADDR	Query Device Bluetooth Address
6	AT+NAME	Query/Set Device Name
7	AT+RNAME	Query Remote Bluetooth Device's Name
8	AT+ROLE	Query/Set Device Role
9	AT+CLASS	Query/Set Class of Device CoD
10	AT+IAC	Query/Set Inquire Access Code
11	AT+INQM	Query/Set Inquire Access Mode
12	AT+PSWD	Query/Set Pairing Passkey
13	AT+UART	Query/Set UART parameter
14	AT+CMODE	Query/Set Connection Mode
15	AT+BIND	Query/Set Binding Bluetooth Address
16	AT+POLAR	Query/Set LED Output Polarity
17	AT+PIO	Set/Reset a User I/O pin
18	AT+MPIO	Set/Reset multiple User I/O pin
19	AT+MPIO?	Query User I/O pin
20	AT+IPSCAN	Query/Set Scanning Parameters
21	AT+SNIFF	Query/Set SNIFF Energy Savings Parameters
22	AT+SENM	Query/Set Security & Encryption Modes
23	AT+RMSAD	Delete Authenticated Device from List
24	AT+FSAD	Find Device from Authenticated Device List
25	AT+ADCN	Query Total Number of Device from Authenticated Device List
26	AT+MRAD	Query Most Recently Used Authenticated Device
27	AT+STATE	Query Current Status of the Device
28	AT+INIT	Initialize SPP Profile
29	AT+INQ	Query Nearby Discoverable Devices
30	AT+INQC	Cancel Search for Discoverable Devices
31	AT+PAIR	Device Pairing
32	AT+LINK	Connect to a Remote Device
33	AT+DISC	Disconnect from a Remote Device
34	AT+ENSNIFF	Enter Energy Saving mode
35	AT+EXSNIFF	Exit Energy Saving mode

ERROR CODES

ERROR CODE	VERBOSE
0	Command Error/Invalid Command
1	Results in default value
2	PSKEY write error
3	Device name is too long (>32 characters)
4	No device name specified (0 lenght)
5	Bluetooth address NAP is too long
6	Bluetooth address UAP is too long
7	Bluetooth address LAP is too long
8	PIO map not specified (0 lenght)
9	Invalid PIO port Number entered
A	Device Class not specified (0 lenght)
B	Device Class too long
C	Inquire Access Code not Specified (0 lenght)
D	Inquire Access Code too long
E	Invalid Iquire Access Code entered
F	Pairing Password not specified (0 lenght)
10	Pairing Password too long (> 16 characters)
11	Invalid Role entered
12	Invalid Baud Rate entered
13	Invalid Stop Bit entered
14	Invalid Parity Bit entered
15	No device in the Pairing List
16	SPP not initialized
17	SPP already initialized
18	Invalid Inquiry Mode
19	Inquiry Timeout occured
1A	Invalid/zero lenght address entered
1B	Invalid Security Mode entered
1C	Invalid Encryption Mode entered

1. Test UART connection

COMMAND	RESPONSE
AT↵	OK↵

2. Reset Device

COMMAND	RESPONSE
AT+RESET↵	OK↵

3. Querry firmware version

COMMAND	RESPONSE
AT+VERSION?↵	+VERSION:<VER>↵ OK↵

where <VER> = Version Number

4. Restore settings to Factory Defaults

COMMAND	RESPONSE
AT+ORGL↵	OK↵

Restore to the following settings:
Device Class: 0
Inquiry Code: 0x009e8b33
Device Mode: Slave
Binding Mode: SPP
UART: 38400bps, 8 bit, 1 stop bit, no parity
Pairing Code: 1234
Device Name: H-C-2010-06-01

5. Query EGBT-045MS Bluetooth Address

COMMAND	RESPONSE
AT+ADDR?↵	+ADDR:nn:uu:ll↵ OK↵

Bluetooth Address Format:
nn - NAP (16 bit Non-significant Address Portion)
uu - UAP (8 bit Upper Address Portion)
ll - LAP (24 bit Lower Address Portion)
Returned bluetooth address

Example: Querry EGBT-045MS Bluetooth Address

From Host controller:
AT+ADDR?↵
EGBT-045MS Response
+ADDR:11:6:230154↵
OK↵

Bluetooth Address is 11:06:23:01:54

6. Query/Set Device Name

COMMAND	RESPONSE
AT+NAME?↵	+NAME:<name>↵ OK↵
AT+NAME=<name>↵	OK↵

where <name> = Device Name (31 characters max.)

Example: Query device name

From Host controller:
AT+NAME?↵
EGBT-045MS Response
+NAME:HC-05↵
OK↵

Example: Set device name to “e-Gizmo”

From Host controller:
AT+NAME=e-Gizmo↵
EGBT-045MS Response
OK↵

Example: Set device name to “supercalifragilisticexpialidocious”

From Host controller:
AT+NAME=supercalifragilisticexpialidocious↵
EGBT-045MS Response
ERROR:(3)↵ *name too long (>31 characters)*

7. Query Remote Bluetooth Device’s Name

COMMAND	RESPONSE
AT+RNAME?<addr>↵	+NAME:<name>↵ OK↵

where <name> = Device name
<addr> = 48 bit bluetooth address
in NAP,UAP,LAP format

Example: Query remote Bluetooth device having address = 00:02:72:0A:3C:7F

Bluetooth address in NA:UAP:LAP format = 0002:72:0A3C7F

From Host controller:
AT+RNAME?0002,72,0A3C7F↵
EGBT-045MS response if remote device name is “HC-05”
+NAME:HC-05↵
OK↵
EGBT-045MS response if remote device name is unresolved
FAIL↵

8. Query/Set Device Role

COMMAND	RESPONSE
AT+ROLE?↵	+ROLE:<role>↵ OK↵
AT+ROLE=<role>↵	OK↵

where <role>
0 - Slave (default)
1 - Master
2 - Slave-Loop

Slave - EGBT-045MS acts as discoverable wireless UART device ready for transparent data exchange.
Master - Scans for a remote bluetooth (slave) device, pairs, and setup connection for a transparent data exchange detween devices
Slave-Loop - Data loop-back Rx-Tx. Used mainly for testing.

Example: Set EGBT-045MS in master role

Bluetooth address in NA:UAP:LAP format = 0002:72:0A3C7F

From Host controller:
AT+ROLE=1↵
EGBT-045MS response
+ROLE:1↵
OK↵

9. Query/Set Class of Device CoD

COMMAND	RESPONSE
AT+CLASS?↵	+CLASS:<class>↵ OK↵
AT+CLASS=<class>↵	OK↵

where <class>
0 - (default)

The Class of Device identifier. For more info, see the Bluetooth_Code_Definition.pdf file included with the product documentation of this kit.

10. Query/Set Inquire Access Code

COMMAND	RESPONSE
AT+IAC?↵	+IAC:<iac>↵ OK↵
AT+IAC=<iac>	OK↵ / FAIL↵

where <iac> = Inquire Access Code
9e8b33 - default value

11. Query/Set Inquire Access Mode

COMMAND	RESPONSE
AT+INQM?↵	+INQM:<inq1>,<inq2>,<inq3>↵ OK↵
AT+INQM=<inq1>,<inq2>,<inq3>↵	OK↵ / FAIL↵

where <inq1> Inquire Access Mode
0 - standard
1 - rssi (default)

<inq2> Maximum number of devices response
0 to 32000
1 (default)

<inq3> Inquire timeout
1 to 48
48 (default)

Maximum number of devices response - EGBT-045MS will stop inquiring once the number of devices that responded reaches this value.
Inquire Timeout - Multiply this number by 1.28 to get the maximum time in seconds the EGBT-045MS will wait for a respond to an inquiry call.

Example: Set EGBT-045MS Inquire Access Mode at
Inquire access mode - 1 (rssi)
Number of devices - 3
Timeout - 10 (10*1.28 = 12.8 seconds)

Bluetooth address in NA:UAP:LAP format = 0002:72:0A3C7F

From Host controller:
AT+INQM=1,3,10↵
EGBT-045MS response
OK↵

12. Query/Set Pairing Passkey

COMMAND	RESPONSE
AT+PSWD?↵	+PSWD:<password>↵ OK↵
AT+PWSD=<password>↵	OK↵

where <password> = Alphanumeric password 16 characters max.
1234 - (default)

Example: Set EGBT-045MS Password to “e-Gizmo”

From Host controller:
AT+PSWD=e-Gizmo↵
EGBT-045MS response
OK↵

13. Query/Set UART parameter

COMMAND	RESPONSE
AT+UART?↵	+UART:<baud>,<stop>,<parity>↵ OK↵
AT+UART=<baud>,<stop>,<parity>↵	OK↵

where <baud> =baud rate, any one of the following
4800
9600 (default)
19200
38400
57600
115200
234000
460800
921600
1382400

<stop> = number of stop bits
0 - 1 bit (default)
1 - 2 bits

<parity> = Parity bit
0 - None (default)
1 - Odd parity
2 - Even Parity

Example: Set EGBT-045MS UART parameter to 115200bps, 2 stop bits, even parity

From Host controller:
AT+UART=115200,1,2↵
EGBT-045MS response
OK↵

14. Query/Set Connection Mode

COMMAND	RESPONSE
AT+CMODE?↵	+CMODE:<mode>↵ OK↵
AT+CMODE=<mode>↵	OK↵

where <mode>
0 - Connect to a specified Bluetooth device only (default). See related command Command 15.
1 - Can connect with any other Bluetooth device.
2 - Test mode

15. Query/Set Binding Bluetooth Address

COMMAND	RESPONSE
AT+BIND?↵	+BIND:<addr>↵ OK↵
AT+BIND=<addr>↵	OK↵

where <addr> = 48 bit bluetooth address in NAP,UAP,LAP format

Example: Bind with Bluetooth device having address = 00:02:72:0A:3C:7F

Bluetooth address in NA,UAP,LAP format = 0002,72,0A3C7F

From Host controller:
AT+BIND=0002,72,0A3C7F↵
EGBT-045MS response
OK↵

16. Query/Set LED Output Polarity

COMMAND	RESPONSE
AT+POLAR?↵	+POLAR:<led1>,<led2>↵ OK↵
AT+POLAR=<led1>,<led2>↵	OK↵

where <led1> = LED1 (pin 31) Polarity
0 - LED1 output active low
1 - LED1 output active high (default)

<led2> = LED2 (pin 32) Polarity
0 - LED2 output active low
1 - LED2 output active high (default)

LED 1
Flashes once each seconds to indicate EGBT-045MS is in Command Mode. Flashes two times per second when EGBT-045MS is in data mode.

LED2
Turns ON when EGBT-045MS remote connection is successfuly opened.

17. Set/Reset a User I/O pin

COMMAND	RESPONSE
AT+PIO=<pn>,<value>↵	OK↵

where <pn> = port number. Available port are as follows
2 - PIO2
3 - PIO3
4 - PIO4
5 - PIO5
6 - PIO6
7 - PIO7
10 - PIO10

<value>
0 - Logic Low
1 - Logic High

Example: Set PIO2 to logic High

From Host controller:
AT+PIO=2,1↵
EGBT-045MS response
OK↵

18. Set/Reset multiple User I/O pin

COMMAND	RESPONSE
AT+MPIO=<iomap>↵	OK↵

where
<iomap> =12-bit I/O map presented in hexadecimal

x	PIO10	X	X	PIO7	PIO6	PIO5	PIO4	PIO3	PIO2	x	x

x - don't care/reserved

Example:
Set PIO2 and PIO6 to logic High, all others to logic 0

Bit pattern is 0000 0100 0100 = 44 hexadecimal

From Host controller:
AT+MPIO=44↵
EGBT-045MS response
OK↵

19. Query User I/O pin

COMMAND	RESPONSE
AT+MPIO?↵	+MPIO:<iomap>↵ OK↵

where
<iomap> =12-bit I/O map presented in hexadecimal

x	PIO10	X	X	PIO7	PIO6	PIO5	PIO4	PIO3	PIO2	x	x

x - reserved -used by system, may assume any values

Example:
Read PIO inputs

From Host controller:
AT+MPIO?↵
EGBT-045MS response
+MPIO:944↵
OK↵
Returned value in binary: 1001 0100 0100

In this example, the PIO are previously set in command 18 with PIO2 and PIO6 set. The returned value also shows reserved bits 11 and 8 set by the system.

20. Query/Set Scanning Parameters

COMMAND	RESPONSE
AT+IPSCAN?↵	+IPSCAN:<int>,<dur>,<pint>,<pdur>↵ OK↵
AT+IPSCAN=<int>,<dur>,<pint>,<pdur>↵	OK↵

where <int> = inquire scan time interval
1024 - default
<dur> = inquire scan time duration
512 - default
<pint> = page scan time interval
1024 - default
<pdur> = page scan time duration
512 - default

All parameters must be represented with decimal integer value.

21. Query/Set SNIFF Energy Savings Parameters

COMMAND	RESPONSE
AT+SNIFF?↵	+SNIFF:<tmax>,<tmin>,<retry>,<timeout>↵ OK↵
AT+SNIFF=<tmax>,<tmin>,<retry>,<timeout>↵	OK↵

where <tmax> = maximum time
0 - default
<tmin> = minimum time
0 - default
<retry> = retry time
0 - default
<timeout> = timeout
0 - default

All parameters must be represented with decimal integer value.

22. Query/Set Security & Encryption Modes

COMMAND	RESPONSE
AT+SENM?↵	+SENM:<mode>,<encrypt>↵ OK↵
AT+SENM=<mode>,<encrypt>↵	OK↵

where <mode> = Security Mode
0 - sec_mode_off (default)
1 - sec_mode1_non-secure
2 - sec_mode2-service
3 - sec_mode3_link
4 - sec_mode_unknown

<encrypt> = encryption mode
0 - hci_enc_mode_off (default)
1 - hci_enc_mode_pt_to_pt
2 - hci_enc_mode_pt_to_pt_and_bcast

23. Delete Authenticated Device from List

COMMAND	RESPONSE
AT+RMSAD=<addr>↵	OK↵

where <addr> = 48 bit bluetooth address
in NAP,UAP,LAP format

Example: Remove from Authenticated Device list a Bluetooth device having address = 00:02:72:0A:3C:7F

Bluetooth address in NA,UAP,LAP format = 0002,72,0A3C7F

From Host controller:
AT+RMASD=0002,72,0A3C7F↵
EGBT-045MS response if deletion is successful
OK↵
EGBT-045MS response if remote device address is not in the list
ERROR(15)↵

Caution:

Entering
AT+RMASD↵
will delete ALL authenticated device from the list!

24. Find Device from Authenticated Device List

COMMAND	RESPONSE
AT+FSAD=<addr>↵	OK↵

where <addr> = 48 bit bluetooth address
in NAP,UAP,LAP format

Note: AT+FSAD returns a FAIL response if device is not in the authenticated list

25. Query Total Number of Device from Authenticated Device List

COMMAND	RESPONSE
AT+ADCN?↵	+ADCN:<total>↵ OK↵

where
<total> = total number of devices in the authenticated device list

26. Query Most Recently Used Authenticated Device

COMMAND	RESPONSE
AT+MRAD?↵	+MRAD:<addr>↵ OK↵

where <addr> = 48 bit bluetooth address
in NAP:UAP:LAP format

27. Query Current Status of the Device

COMMAND	RESPONSE
AT+STATE?↵	+STATE:<stat>↵ OK↵

where
<stat> = Current Status, any one of the following:

INITIALIZED
READY
PAIRABLE
PAIRED
INQUIRING
CONNECTING
CONNECTED
DISCONNECTED
UNKNOWN

28. Initialize SPP Profile

COMMAND	RESPONSE
AT+INIT↵	OK↵ / FAIL↵

29. Query Nearby Discoverable Devices

COMMAND	RESPONSE
AT+INQ↵	+INQ: <addr>,<class>,>rssi>↵ OK↵

where <addr> = 48 bit bluetooth address
in NAP:UAP:LAP format
<class> = Device Class
<rssi> = RSSI

This command will scan and report all nearby dicoverable Bluetooth devices. The same device may be reported more than once.

This command will work only if EGBT-045MS is set to work as a master device, i.e. AT+ROLE=1, and after AT+INIT is executed.

Example: Discover nearby devices

From Host controller: Set device role as master

AT+ROLE=1↵

EGBT-045MS response

OK↵

From Host controller: Initialize SPP

AT+INIT↵

EGBT-045MS response

OK↵

From Host controller: Set inquire mode as RSSI, look for 9 devices, and 48 as timeout

AT+INQM=1,9,48↵

EGBT-045MS response

OK↵

From Host controller: Start Device Discovery

AT+INQ↵

EGBT-045MS response (sample only, actual report will vary)

+INQ:101D:C0:2E7B54,5A0204,7FFF↵

+INQ:25:48:21AD1A,5A020C,7FFF↵

OK↵

In this example, EGBT-045MS found only two discoverable devices. It will quit searching after the timeout time specified by the AT+INQM command or if the number of discovered devices equals the number of specified devices.

30. Cancel Search for Discoverable Devices

COMMAND	RESPONSE
AT+INQC↵	OK↵

AT+INQ can be stopped at anytime by executing this command.

31. Device Pairing

COMMAND	RESPONSE
AT+PAIR=<addr>,<timeout>↵	OK↵ / FAIL↵

where <addr> = 48 bit bluetooth address in NAP,UAP,LAP format

<timeout> = Timeout time in sec

32. Connect to a Remote Device

COMMAND	RESPONSE
AT+LINK=<addr>↵	OK↵ / FAIL↵

where <addr> = 48 bit bluetooth address in NAP,UAP,LAP format

33. Disconnect from a Remote Device

COMMAND	RESPONSE
AT+DISC↵	+DISC:<results>↵

where

<results> = Disconnection results, any one of the following:

SUCCESS
LINK_LOSS
NO_SLC
TIMEOUT
ERROR

34. Enter Energy Saving mode

COMMAND	RESPONSE
AT+ENSNIFF=<addr>↵	OK ↵

where <addr> = 48 bit bluetooth address in NAP,UAP,LAP format

35. Exit Energy Saving mode

COMMAND	RESPONSE
AT+EXSNIFF=<addr>↵	OK↵

where <addr> = 48 bit bluetooth address in NAP,UAP,LAP format